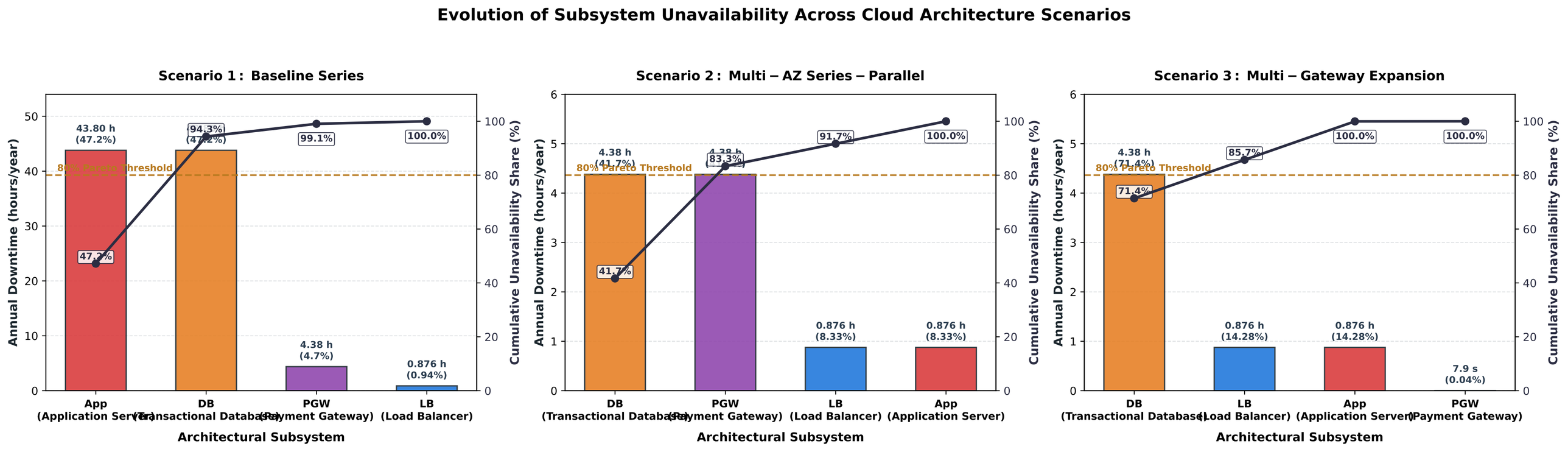
Evolution of Subsystem Unavailability Across Cloud Architecture Scenarios
Scenario 1 : Baseline Series
Annual Downtime (hours/year)
Cumulative Unavailability Share (%)
80% Pareto Threshold
43.80 h
(47.2%)
94.3%
99.1%
100.0%
47.2%
4.38 h
(4.7%)
0.876 h
(0.94%)
App
(Application Server)
DB
(Transactional Database)
PGW
(Payment Gateway)
LB
(Load Balancer)
Architectural Subsystem
Scenario 2 : Multi – AZ Series – Parallel
Annual Downtime (hours/year)
Cumulative Unavailability Share (%)
80% Pareto Threshold
4.38 h
(41.7%)
4.38 h
(41.7%)
83.3%
91.7%
100.0%
41.7%
0.876 h
(8.33%)
0.876 h
(8.33%)
DB
(Transactional Database)
PGW
(Payment Gateway)
LB
(Load Balancer)
App
(Application Server)
Architectural Subsystem
Scenario 3 : Multi – Gateway Expansion
Annual Downtime (hours/year)
Cumulative Unavailability Share (%)
80% Pareto Threshold
4.38 h
(71.4%)
85.7%
100.0%
100.0%
71.4%
0.876 h
(14.28%)
0.876 h
(14.28%)
7.9 s
(0.04%)
DB
(Transactional Database)
LB
(Load Balancer)
App
(Application Server)
PGW
(Payment Gateway)
Architectural Subsystem